

HAMIDREZA KAMKARI

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<https://hamidrezakmk.github.io/>

EDUCATION

Massachusetts Institute of Technology (MIT)

2025 - 2031 (Ongoing)

Doctor of Philosophy Student

Research lies at the intersection of high-dimensional geometry, machine learning, and simulation. Working under the supervision of Prof. Justin Solomon, in the geometric data processing lab.

University of Toronto

2022 - 2024

Master of Science

Mitacs Accelerate Fellowship

Department of Computer Science

Cumulative GPA (4.0/4)

Courses: (CSC2421) Graphs, Matrices, and Optimization, (CSC2541) Topics in Machine Learning: Introduction to Causality

Sharif University of Technology

2018 - 2022

Bachelor of Science

Overall GPA 19.22/20

Department of Computer Engineering

Ranked Among the Top 10

Courses: (CE695) Stochastic Processes, (CE242) Signals & Systems, (CE282) Linear Algebra, (CE181) Probability and Statistics, (CSC384) Artificial Intelligence, (CE354) Algorithm Design, (CE415) Formal Languages, (MAT034) Differential Equations

PUBLICATIONS

- Learning Orthonormal Bases for Function Spaces
Hamidreza Kamkari, Mohammad Sina Nabizadeh, Justin Solomon
Preprint: [arXiv: 2605.19959](#)
- IV-ICL: Bounding Causal Effects with Instrumental Variables via In-Context Learning
Vahid Balazadeh, **Hamidreza Kamkari**, Medha Barath, Ricardo Silva, Rahul G. Krishnan
Preprint: [arXiv: 2605.12924](#)
- CausalPFN: Amortized Causal Effect Estimation via In-Context Learning
Vahid Balazadeh*, **Hamidreza Kamkari***, Valentin Thomas, Benson Li, Junwei Ma, Jesse C. Cresswell, Rahul G. Krishnan
NeurIPS 2025 (**Spotlight**): [arXiv: 2506.07918](#)
Accepted for an *oral* presentation at the SIM workshop at ICML 2025
- A Geometric View of Data Complexity: Efficient Local Intrinsic Dimension Estimation with Diffusion Models
Hamidreza Kamkari, Brendan Ross, Rasa Hosseinzadeh, Jesse Cresswell, Gabriel Loaiza-Ganem
NeurIPS 2024 (**Spotlight**): [arXiv: 2406.03537](#)
Also accepted to three ICML 2024 workshops with two contributed talks and spotlight presentations
- A Geometric Explanation of the Likelihood OOD Detection Paradox
Hamidreza Kamkari, Brendan Ross, Jesse Cresswell, Anthony Caterini, Rahul Krishnan, Gabriel Loaiza-Ganem
ICML 2024 (Poster): [arXiv: 2403.18910](#)
- A Geometric Framework for Understanding Memorization in Generative Models
Brendan Ross, **Hamidreza Kamkari**, Zhaoyan Liu, Tongzi Wu, George Stein, Gabriel Loaiza-Ganem, Jesse C. Cresswell
ICLR 2025 (**Spotlight**): [arXiv: 2411.00113](#)
Also accepted to two ICML 2024 workshops
- TabDPT: Scaling Tabular Foundation Models
Junwei Ma*, Valentin Thomas*, Rasa Hosseinzadeh, **Hamidreza Kamkari**, Alex Labach, Jesse C. Cresswell, Keyvan Golestan, Guangwei Yu, Maksims Volkovs, Anthony L. Caterini
NeurIPS 2025: [arXiv: 2410.18164](#)

WORK EXPERIENCE

Layer 6 AI Lab at Toronto Dominion (TD) Bank
Research Machine Learning Scientist (Full Time)

Toronto, Canada
February 2024 - June 2025

- Led multiple publications in top-tier conferences on topics such as deep generative modeling and geometric deep learning.
- Developed a tabular foundation model for Canada's largest bank utilizing prior fitted networks and a Bayesian approach

Layer 6 AI Lab at Toronto Dominion (TD) Bank
Machine Learning Research Intern

Toronto, Canada
May 2023 - December 2023

- Developed novel out-of-distribution detection methods using deep generative models, leading a first-author publication at a top-tier machine learning conference.

Aalto University
Undergraduate Research Intern

Espoo, Finland
July 2021 - September 2021

- Designed a reinforcement learning pipeline to design complex RNA structures, including previously underexplored pseudo-knotted structures

HONOURS & AWARDS

- Secured **30,000 CAD** in funding for graduate research through the Mitacs Accelerate Scholarship May 2023
- **Gold Medal** in regional ACM-ICPC contests December 2018

TEACHING & SUPERVISION

Vector Institute
Research Mentoring

Toronto, Canada
July 2024 - Ongoing

- Started as a researcher at the Vector Institute under the supervision of Prof. Rahul Krishnan, and currently helping mentor undergraduate students Aidan Li and Benson Li, both of whom are affiliated with his lab

University of Toronto
Teaching Assistance

Toronto, Canada
September 2022 - May 2023

- **CSC384**: Introduction to Artificial Intelligence — **CSC236**: Introduction to the Theory of Computation

Sharif University of Technology
Teaching Assistance

Tehran, Iran
September 2019 - July 2022

- **CE40254**: Data Structure and Algorithms (*Head TA*) — **CE40181**: Probability and Statistics — **CE40417**: Artificial Intelligence

PEER-REVIEW & TALKS

- SIAM Journal on Imaging Sciences (SIIMs) reviewer Annual (Ongoing)
- Invited Talk at Visual Computing (VC) Seminar MIT 2025
- Advancements in Neural Information Processing Systems Reviewer NeurIPS 2023-2025
- Talk on Generative Modelling through the Lens of Manifold Hypothesis (Remote) Imperial College London
- Association for the Advancement of Artificial Intelligence Program Committee AAAI 2025
- International Conference on Learning Representations Reviewer ICLR 2024/2025
- International Conference in Machine Learning Reviewer ICML 2024
- Transactions on Machine Learning Research (TMLR) Reviewer Annual (Ongoing)
- **Contributed Talk** for workshop: Differentiable Almost Everything ICML 2024
- **Contributed Talk** for workshop: Structured Probabilistic Inference and Generative Modelling ICML 2024

ADDITIONAL RESEARCH EXPERIENCE

Max Planck Institute for Informatics
Undergraduate Research Intern

Saarbrücken, Germany
January 2022 - March 2022

- Physarum Inspired Dynamics to Solve Semi-Definite Programs
Yuan Gao, *Hamidreza Kamkari*, Andreas Karrenbauer, Kurt Mehlhorn, Mohammadamin Sharifi
Pre-print from the internship: [arXiv: 2111.02291](https://arxiv.org/abs/2111.02291)